

LECTURE 16: MATH1061/7861
TOPICS: RECURSIVE DEFINITIONS
DATE: TUESDAY MARCH 31, 2026

Learning objectives:

- (1) Learn recursive definitions of sequences and sets.
- (2) Reinforce mathematical induction.

* There were issues with lecture recording this past Thursday. We have notified the relevant people.

* No tutorials on Friday. Ensure you go to earlier ones.

Q1:

Let S be the subset of all binary strings, defined as follows:

Base: The empty string is in S .

Recursion: If $s \in S$ then $s1 \in S$ and $s01 \in S$.

Restriction: Nothing is in S other than the objects defined by the recursion.

Example: $e = \text{empty string} \in S$
 $e1 = 1 \in S$, $01 \in S$
 $\Downarrow$
 $11 \in S$, $101 \in S$

(a) Is $1101 \in S$? Explain Yes. We have already seen $11 \in S$. Recursion Rule says $1101 \in S$.

(b) Is $0110 \in S$? Explain No. Recursion specifies that we can only ever end in 1.

(c) State a necessary property for a string to be in S .

(i.e. a property all strings in S must have)

Is any
of this
sufficient?

★ It's either empty string or ends in 1.

Not sufficient
e.g. 001
↓

★ Cannot contain 00. Not sufficient 10

★ must be either empty or contain
1's ~~and~~ or 01's. Not sufficient 0110

★ $\#1's \geq \#0's$ Not sufficient,

(d) Describe strings of length 6 in S without explicitly writing them down, but using lower length strings.

How many strings of length 6 are in S ?

$F_n = \#$ of strings in S of length n .

We want to know F_6

What is F_6 in terms

$F_5, F_4, \dots$

$$F_n = F_{n-1} + F_{n-2}$$

Fibonacci

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(e) Write all strings in S of length 0, 1, 2, 3, 4, 5.

1 length 0 : ϵ ← empty string

1 length 1 :

2 length 2 :

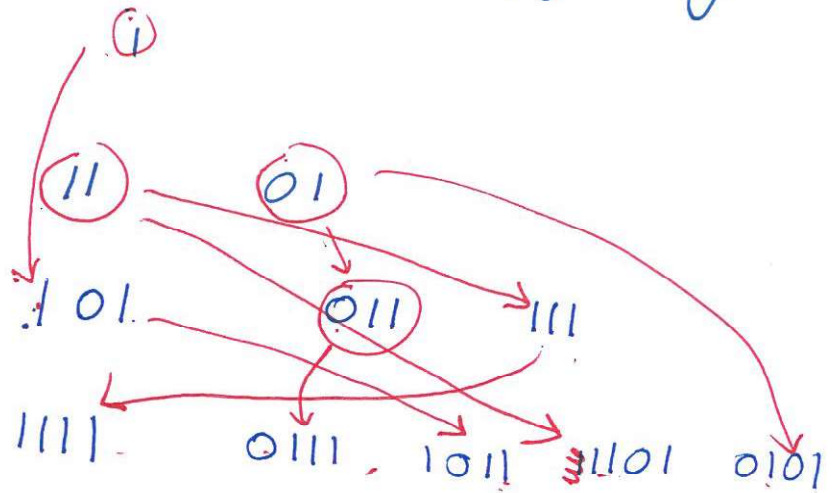
3 length 3 :

5 length 4 :

8 length 5 :

13 length 6 :

21 length 7 :



- (f) Let F_n be the number of strings of length n in S . Give a recursive formula for F_n .

$$F_n = F_{n-1} + F_{n-2}$$

- (g) Challenge: Give a necessary and sufficient condition for a string to be in S .

Not assessable